



MMAUG / AI & DEVOPS BOOTCAMP

AZURE FUNDAMENTALS FOR BEGINNERS

Mapping Cloud Services to Your First AI App Idea

| Arinze Edeh / Cloud & DevOps Engineer

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Meet the speaker

Self-taught and building in public, with a focus on cloud and DevOps



Arinze Edeh

Cloud & DevOps Engineer

Arinze is a self-taught Cloud and DevOps Engineer with a public GitHub portfolio of 200+ documented infrastructure implementations. He holds a top-10 global ranking on KodeKloud's Senior DevOps Engineer leaderboard and is completing a B.Sc. in Computer Science with a 4.0 GPA.

- AZ-104 Microsoft Certified, AZ-900 Microsoft Certified, IBM Cybersecurity Analyst Professional Certificate
- Volunteer speaker for MMAUG; builds in public with a 200+ implementation GitHub portfolio
- Believes the hardest part of building with AI isn't the model, it's knowing which cloud service to reach for first

CONNECT

github.com/arinze-edeh



Today's session

From a vague idea to a working Azure AI call

01 — Context

Beginners have an AI app idea but freeze in front of Azure's full service catalog

02 — Concepts

The six categories every cloud app needs, and where Azure AI services fit

03 — Practice

Map your own AI app idea to the six categories (no account needed)

04 — Demo

One live Azure AI call, shown start to finish

05 — Next steps

Optional guided mini-lab, free credits, learning paths and the starter repo



What participants will learn

Four things you will be able to do by the end

01

01

Explain

The core Azure categories beginners actually need for an AI app

02

02

Design

A simple architecture mapping an AI app idea onto Azure services

03

03

Build

A first working call to an Azure AI service (hands-on in the optional lab)

04

04

Evaluate

Which Azure service fits a requirement, and which is overkill

Each outcome is practised in the mapping exercise or the live demo



SECTION / 01

FROM IDEA TO ARCHITECTURE

Turning a vague AI app idea into a concrete set of Azure services





What Is an Azure Service?

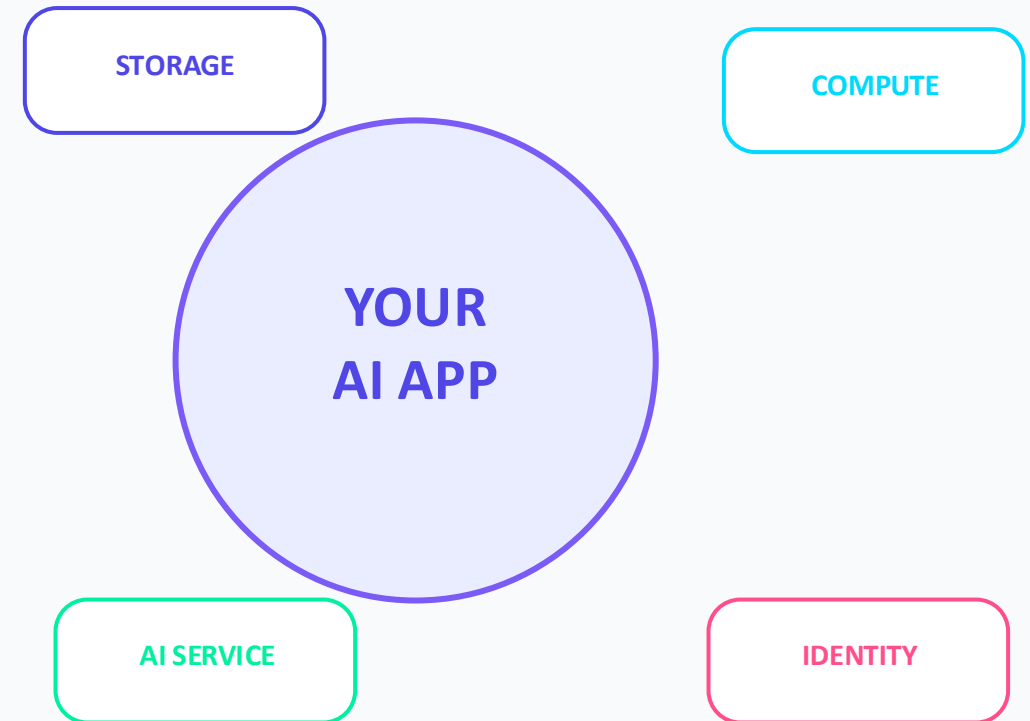
Every AI app is built from the same core parts

A managed building block that Azure runs for you, so you don't operate the underlying servers yourself

Every AI app is really just compute, storage, intelligence and identity wired together (the next slide adds the rest). Once you can name which Azure service plays each role, the platform stops feeling overwhelming.

EXAMPLE

Upload a photo, get a caption back: that's Blob Storage, an Azure Function and Azure AI Vision working together





The six categories

What every cloud app needs, and an Azure service for each

01

01

COMPUTE

Functions, App Service, VMs

02

02

STORAGE

Blob Storage, Cosmos DB

03

03

AI SERVICES

Azure Vision, Foundry Tools

04

04

NETWORKING

Virtual Network, HTTPS

05

05

IDENTITY

Microsoft Entra ID

06

06

MONITORING

Azure Monitor + App Insights



IaaS (Virtual Machines) and PaaS (App Service)

Two ways to host an app: choose by how much control you need

IaaS: VIRTUAL MACHINES

- Full control over OS and runtime
- Maximum flexibility and control
- You manage patching, scaling and security
- Custom legacy workloads or specific OS needs

PaaS: APP SERVICE

- Fast deployment for standard web/AI apps
- No server management, built-in scaling
- Less control over the underlying OS
- A beginner's first AI-powered web app

DECISION

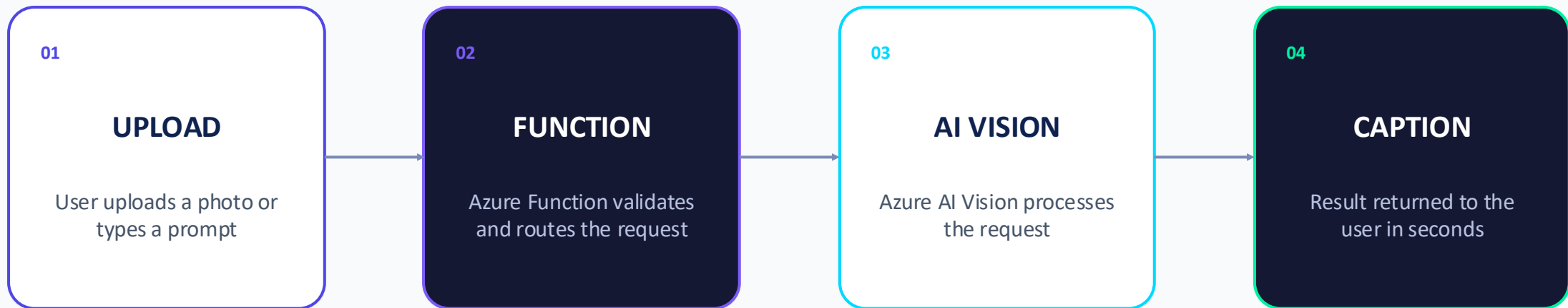
Choose App Service unless your workload needs OS-level control





A First AI App on Azure

One request, four responsibilities



Identity via Microsoft Entra ID, secrets in Key Vault, monitoring via Azure Monitor



Map your idea to Azure

No Azure account needed: 25 minutes with the worksheet

TIMEBOX

25

MINUTES

WORK MODE

Individual or pairs

PREREQUISITE

None, just your idea

YOUR TASK

Break your own AI app idea into parts and match each part to an Azure category

- Write your idea in one sentence
- List its input, process, intelligence and output
- Match each part to a category: compute, storage, identity, networking, monitoring, AI
- Ask: could a simpler service do the job?

SUCCESS CHECK

A filled table with one Azure service per role ([docs/idea-to-services-worksheet.md](#))





Live demo walkthrough

Success: a real caption printed in the terminal

DEMO GOAL

Call an Azure AI Vision endpoint from a simple script and get back a real result

- 1 Sign in to the Azure Portal and create an AI Vision resource
- 2 Copy the endpoint and key into a .env file
- 3 Run the script against a sample image
- 4 Confirm the returned description matches the image

```
$ python src/vision_demo.py --tags
Caption: "a woman wearing a mask sitting at a table with a laptop" (confidence 0.85)
Tags:
furniture (0.99)
clothing (0.98)
person (0.94)
houseplant (0.94)
desk (0.92)
indoor (0.90)
laptop (0.88)
computer (0.85)
sitting (0.81)
wall (0.75)
woman (0.74)
table (0.68)
plant (0.64)
using (0.54)
```

Fallback: recorded demo + starter repo, linked in Resources



Optional guided mini-lab

For anyone with a free Azure account: 25 minutes, one clear result

TIMEBOX

25

MINUTES

WORK MODE

Individual, optional

PREREQUISITE

Free Azure account
(optional)

YOUR TASK

Create an Azure AI Vision resource and get a live description back for your own image

- Sign in to portal.azure.com
- Create a Computer Vision resource (free tier)
- Clone the repo, install packages, add your endpoint and key to `.env`
- Run it against any image and read the output

SUCCESS CHECK

A caption printed in your terminal; finish at home if you need more time





Code walkthrough

A handful of lines do the work

`vision_demo.py`

```
01 from azure.ai.vision.imageanalysis import ImageAnalysisClient
02 client = ImageAnalysisClient(endpoint, AzureKeyCredential(key))
03 result = client.analyze_from_url(url, [VisualFeatures.CAPTION])
04 print(result.caption.text)
05 # endpoint and key load from .env, never hardcoded
06 $ python src/vision_demo.py
```

[1]

Only three lines actually talk to Azure, everything else is plumbing

[2]

The key comes from your .env file, never hardcoded in the script

[3]

The terminal prints a one-sentence caption describing your image



Key practices and guardrails

Connect technical choices to reliability, security and responsible use

01

01

Security

Managed Identity instead of hardcoded keys

02

02

Quality

Test the AI call against several sample images before demoing live

03

03

Operations

Basic alerts in Azure Monitor so failures don't go unnoticed

04

04

Responsible AI

Review Azure's Responsible AI guidelines before shipping to real users

The starter repo keeps keys in .env and includes a key-free Entra ID stretch task





Continue learning

Everything from today, in one place

SLIDES

This deck, shared in the session chat

[Session chat](#)

CODE

Starter repo with the vision_demo.py sample

[arinze-edeh/azure-fundamentals-first-ai-app](#)

DOCS

Microsoft Learn: Azure AI Vision

learn.microsoft.com/azure

COMMUNITY

MMAUG events and updates

mmaug.com



Scan for the starter repo

Meetup: meetup.com/malta-microsoft-ai-user-group



THANK YOU.

Thank you for joining. If you build something from today's session, I'd love to see it.

WEBSITE

mmaug.com

MEETUP

Malta Microsoft AI User Group

SPEAKER

github.com/arinze-edeh

